Foundational Skills Demands Across Disciplines: A Comparative Analysis Between STEM and BHASE Co-op Job Postings

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Abstract

The rapid advancement of Large Language Models (LLMs) and Generative AI is transforming the modern workplace, increasing the importance of foundational skills that help students to adapt to evolving professional demands. This study analyzes cooperative education job postings advertised to undergraduate students at a mid-sized university at Southern Ontario between 2012 and 2025 to identify the foundational skills most sought by employers and to examine how these demands vary across academic programs. Using natural language processing (NLP) techniques, including tokenization, Word2Vec embeddings, BERT-based contextual representations, and a hybrid similarity scoring approach, foundational skills were extracted and ranked according to their relevance within each program. The dataset includes over 478,000 cleaned postings across 75 undergraduate programs. Results indicate that Interpersonal Communication consistently ranks as the most in-demand skill across all fields, followed by Active Listening, Decision Making, Creativity, and Ethics. While foundational skills are broadly consistent across disciplines, minor differences in ranking mainly appear among the lower-ranked skills in the top 20. These variations suggest program-specific priorities, with Science, Technology, Engineering, and Mathematics (STEM) programs showing slightly greater emphasis on analytical reasoning, while Business, Humanities, Arts, Social Sciences, and Education (BHASE) fields place somewhat more emphasis on self-organization. These findings demonstrate that foundational skills are both measurable and stable across disciplines yet subtly shaped by field-specific expectations. The study contributes to labor-market analytics by combining semantic modeling and hybrid embeddings to capture the diverse expressions used in job postings, providing insights for students, educators, and curriculum planning.

Keywords: Foundational Skills, Transferable, NLP, Word2Vec, BERT, STEM, BHASE, Cooperative Education Job Postings.

1. Introduction

Success in today's work environment requires far more than just knowledge gained in the classroom. Foundational or transferable skills such as communication, teamwork, adaptability, and problem-solving are competencies that can be applied across disciplines, industries, and job roles. With the rapid growth of Large Language Models (LLMs) and Generative and autonomous AI, technical tasks are increasingly automated, elevating the importance of human-centered capabilities that machines cannot easily replace. As job roles change quickly, employers are placing greater emphasis on skills related to both intrapersonal and interpersonal dimensions (Fraeys de Veubeke et al.,

2025). For universities, this shift creates an urgent need to align curricula with real labor-market demands to ensure students are prepared for future professional and research environments.

Researchers have studied job postings to better understand what skills employers demand. Earlier studies mainly used simple text analysis methods, such as Term Frequency-Inverse Document Frequency (Qaiser et al.; Liang et al.), Bag-of-Words (Juluru et al.; Qader et al.), and topic modeling (Blei, Ng, and Jordan), to find the most common skill-related words in job descriptions. These approaches focus on counting how often certain terms appear but are not context aware. For instance, the phrases “communication skills are required for this role” and “communication is not required for this role” may be interpreted similarly because both contain the word “communication.” As a result, important contextual differences may be overlooked.

More recent research has used advanced Natural Language Processing (NLP) models that better understand context. For example, Akkasi (2024) developed a method using transformer-based ensemble models to extract both technical and non-technical skills from job postings. The approach improves accuracy while also keeping the results explainable, which is important when using AI tools for real-world analysis. However, this study focused mainly on computer science and IT jobs using postings from several European countries, including the United Kingdom, the Netherlands, France, Ireland, Germany, Scotland, and Belgium. Because of this, it is unclear whether the same methods would give similar results to other fields or in other regions. This suggests a need for further research using context-aware methods on more diverse datasets and academic programs.

Another important study by Khaouja, Mezzour, Carley, and Kassou (2019) introduced TaxoSoft, a method for building a foundational-skill taxonomy using large collections of job ads in English and French. Their approach combined Word2Vec, DBpedia, and network analysis to identify and organize foundational skills as they appear in job postings. They found that foundational kills are difficult to detect using simple keyword searches because they are often written in different forms, such as “teamwork”, “collaboration”, or “working with others.” Their method achieved much higher accuracy than earlier approaches, demonstrating that capturing word and phrase context is more effective than simply counting term frequency and is essential for accurately identifying skill variations in real-world job data. However, this study is based on job ads from American and Moroccan job markets rather than structured co-op job portals that directly connect students with employers, which may limit its applicability to student-focused recruitment contexts.

In Canada, Fraeys de Veubeke and colleagues (2025) studied foundational skills in co-op job postings for engineering and IT students. They analyzed over 75,000 job postings between 2012 and 2023 using Term Frequency-Inverse Document Frequency (TF-IDF). Their results showed that engineering job postings often focused on problem-solving and teamwork, while IT postings highlighted adaptability and communication. They also found that IT job postings usually include more unique foundational skills than engineering postings. Over time, the number of foundational skills listed in job postings increased, suggesting that employers expected more from students than before. This study shows that foundational skills can be measured using job posting data and that employer expectations are changing over time. While the study demonstrates that foundational skills can be analyzed using job posting data, it also has limitations. TF-IDF does not account for the contextual meaning of words, which may limit the accuracy of identifying skills in different contexts. In addition, the study focuses only on engineering and IT programs, leaving uncertainty about how foundational skill demands may differ across other academic disciplines. These limitations highlight the need for further research using context-aware methods and a broader range of academic programs.

Despite the progress and contributions of previous studies, several important gaps remain in the literature. Many existing studies—such as those by Qaiser and Ali, Khaouja et al., Liang and Niu, Blei et al., Jatnika et al., Juluru et al., and Lukauskas et al. — rely on international job boards or general text corpora rather than structured co-op job portals that directly connect students with employers. In addition, while some research compares engineering and IT programs, such as the work by Fraeys de Veubeke et al. (2025), fewer studies examine broader academic groupings such as Science, Technology, Engineering, and Mathematics (STEM) and (Business, Humanities, Arts, Social Sciences, and Education (BHASE). Furthermore, although TF-IDF is widely used to identify important skills, fewer studies combine it with more advanced NLP models such as Word2Vec and BERT to analyze skill patterns across multiple programs over long periods of time. Finally, there is a lack of long-term research based on Canadian co-op job posting data, highlighting the need for more comprehensive and context-specific analysis.

To address these gaps, this thesis analyzes cooperative education job postings from a mid-sized university in Ontario, Canada. The dataset covers the years 2012 to 2025 and includes 478,628 cleaned rows of job-description text across 75 academic programs. Because the dataset is large and unstructured, natural language processing (NLP) methods are used to analyze it efficiently. This study uses Word2Vec to group similar skill expressions together and BERT to better understand the context of skill-related terms. By combining these approaches, the study aims to improve how foundational skills are identified in job postings.

This research has three main goals. First, it aims to identify the most required foundational skills in co-op job posting. While previous studies have examined skills in general job advertisements, fewer studies have analyzed skills using structured co-op job portals that directly connect students with employers. Second, it examines whether skill demands differ between STEM and BHASE programs. Third, it provides insights that can help universities improve curriculum design and better prepare students for the job market and academic research environments. By using modern NLP techniques and long-term Canadian co-op data, this study offers a broader and more detailed understanding of how employer expectations are changing in today's AI-driven economy.

2. Methodology

Data were obtained from the Cooperative Job Portal at a mid-sized university in Southern Ontario, Canada and is provided in two versions, each stored in an Excel file. Because this analysis focuses on program-specific patterns, The ByProgram dataset file was used, which expands the information so that each program gets its own row so that jobs show up multiple times and sorted by programs. This dataset contains 514,446 rows and 31 columns, covering 85 unique undergraduate programs at the mid-size university and tracks the co-op job postings from August 15, 2012, to September 19, 2025.

To prepare the dataset for program-level analysis, the raw ByProgram dataset was cleaned using two main criteria:

a. Remove the EEAPC program
   The program "EEAPC – Environmental Economics & Policy" was excluded from the analysis. Although the program has very few students enrolled, it appeared with the highest program count in the dataset. Because this inconsistency didn't reflect actual student representation, all rows associated with EEAPC were removed from the dataset.

b. Remove all 'inactive' programs
   Some program rows contained the keywords 'inactive', indicating programs that were discontinued or not actively used in the job posting process. Records for inactive programs (identified by the word 'inactive' in the Program column) were removed.

The dataset was loaded into a single Pandas (v2.3.3) data frame, after which the Program column was processed to calculate the frequency of each program. For every

program, the name, count, and percentage of job postings were computed and presented, providing an overview of program representation across the dataset.

The “Job Description” column, which contains text from job postings, was preprocessed using tokenization. This process splits sentences into individual words or meaningful phrases for further analysis. A Python (v.12.9) class was implemented to build the vocabulary, generate token sequences, and extract skill terms along with their alternative labels based on the foundational skills taxonomy proposed by Khaouja, Mane, et al. (2019). To improve corpus quality, sub-sampling was applied to remove extremely high-frequency tokens, reduce noise, and produce more balanced training sequences.

Next, a word embedding model was trained to generate dense vector representations of skill-related terms using the skip-gram architecture of Word2Vec. The model was implemented with TensorFlow Keras (v2.19.0), following the continuous skip-gram approach described by Word2vec model analysis for semantic similarities in English words and the official TensorFlow tutorial. In this architecture, two separate embedding layers were used to represent target and context words. Training data were generated by constructing skip-gram pairs from tokenized job posting text, where each target word was paired with its surrounding context words within a fixed window. To control computational complexity and memory usage, each sequence was limited to a maximum of five skip-gram pairs.

Negative sampling was applied to improve training efficiency and representation quality by approximating the full softmax objective. Specifically, for each positive target–context pair, several negative samples were drawn randomly from the vocabulary distribution, allowing the model to distinguish meaningful word associations from noise. This technique reduces computational cost while preserving semantic learning, as demonstrated in prior work (Jatnika et al., 2019). Through this process, the model learned dense, low-dimensional embeddings in which semantically related skill terms are positioned closer together in vector space, enabling effective similarity computation.

In parallel, a pre-trained BERT model was loaded from TensorFlow Hub (v0.16.1) and fine-tuned on the domain-specific corpus of job postings. Unlike static embeddings such as Word2Vec, BERT produces contextualized representations that capture the meaning of words based on their surrounding context within a sentence. Fine-tuning allowed the model to adapt to domain-specific language patterns, such as technical skills and job-related terminology, improving downstream performance in skill extraction and classification tasks.

To measure the relevance between academic programs and required skill sets, the Python (v12.9), class integrated both static Word2Vec embeddings and contextual

BERT representations. This hybrid approach leverages the complementary strengths of the two models: Word2Vec captures global semantic similarity, while BERT encodes contextual nuance. Following the taxonomy and methodology proposed by Khaouja et al., similarity was computed between job postings and a predefined set of skill terms, including their alternative labels and synonyms.

For each program, a hybrid representation was constructed by concatenating the projected BERT embedding (obtained from the fine-tuned model) with the averaged Word2Vec vector derived from the corresponding job posting text. This combined vector captures both contextual and distributional semantics. Cosine similarity was then used as the similarity metric to compare program vectors with skill term vectors, as it is well-suited for measuring angular similarity in high-dimensional embedding spaces. Skills with similarity scores greater than 0.5 were retained, producing a ranked list of relevant skills for each program.

The results were aggregated and exported to an Excel file, which included program names, extracted skill terms, and their associated similarity scores. This output provides an interpretable mapping between academic programs and industry-relevant skills, supporting downstream analysis and decision-making.

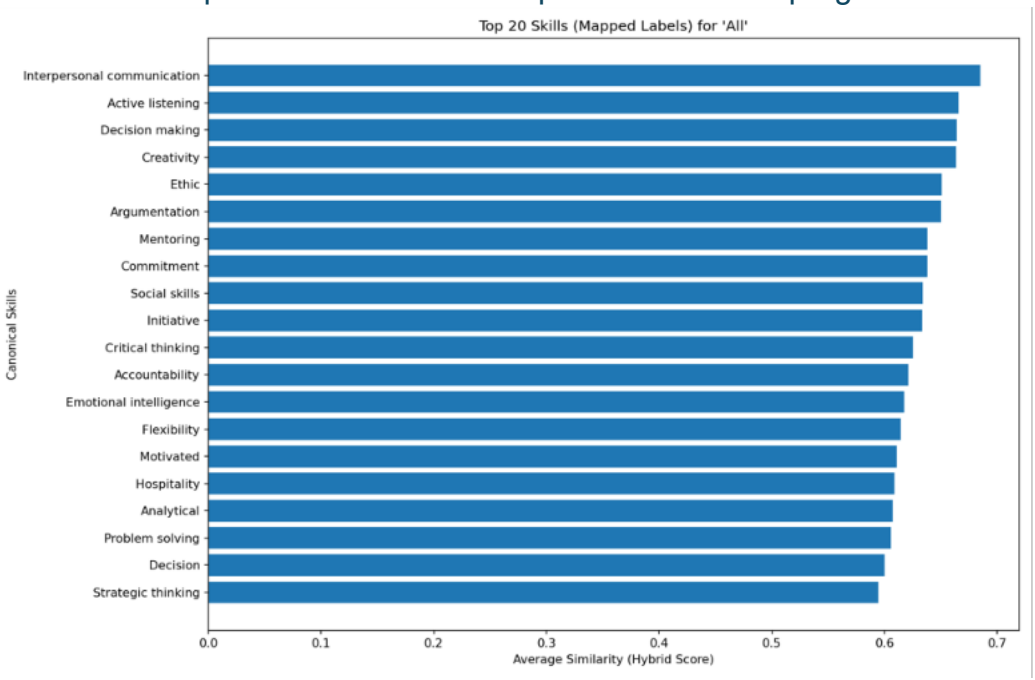
Finally, visualization was performed using Python (v12.9), which processed the Excel output. Skill frequencies were aggregated and sorted by programs based on similarity rankings. The top 20 foundational skills were identified for each program keyword and grouped into program categories according to predefined lists. Histogram plots were generated using Matplotlib to illustrate the distribution of skills across programs and categories, highlighting key foundational skills and differences in emphasis between BHASE and STEM fields.

3. Results

After data cleaning process, 35,818 rows in the original dataset were removed from original 514,446 rows, leaving 478,628 rows covering the years 2012 to 2025 across 75 academic programs.

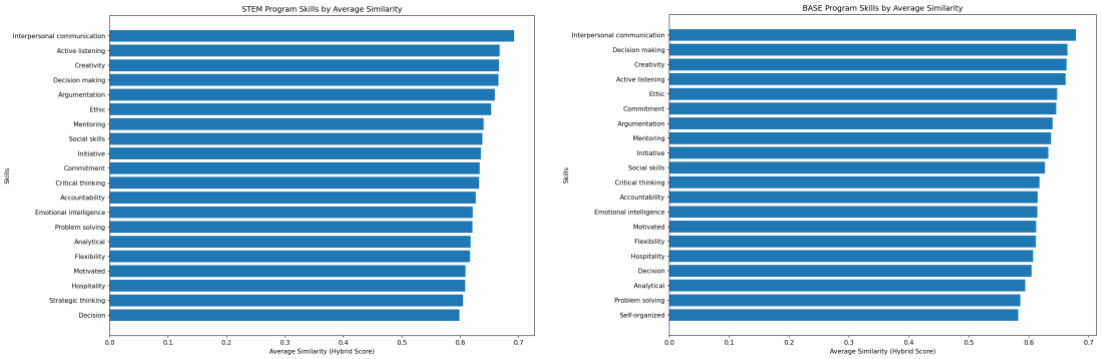
After analysing data by Natural Language Processing (NLP) methods, the top 20 skills are then identified and ranked based on their cosine similarity scores greater than 0.5. The highest-ranked skills include interpersonal communication, active listening, decision-making, creativity, and ethics, as shown in the diagram below.

Top 20 skills in Job Descriptions across the programs



Top 20 skills Across 75 undergraduate programs by Cosine Similarity

For STEM programs, the highest-ranked skills include interpersonal communication, active listening, creativity, decision-making, argumentation, and ethics. In comparison, BHASE programs highlight interpersonal communication, decision-making, creativity, active listening, and ethics as the most prominent skills, as illustrated in the diagrams below.



Top 20 skills for STEM-related programs versus BHASE-related programs

4. Discussion

The analysis of foundational skills in co-op job descriptions across 75 programs at the mid-size university reveals that Interpersonal Communication is the most frequently identified skill, consistently ranking first across all program groups. The next four most common skills are Active Listening, Decision-making, Creativity, and Ethics. The average cosine similarity scores for the top 20 foundational skills generally range between 0.6 and 0.7, indicating strong alignment with the established skills taxonomy. Across multiple runs of the analysis code, the results show minor variations in the ordering of skills, but the overall set of top skills demonstrate high consistency. (See Appendix, Figure 1)

A clearer distinction reveals when comparing STEM and BHASE programs. For STEM programs, the highest-ranked skills include interpersonal communication, active listening, creativity, decision-making, argumentation, and ethics, reflecting a balance between communication and analytical reasoning skills. In contrast, BHASE programs emphasize interpersonal communication, decision-making, creativity, active listening, and ethics, with a relatively stronger focus on organizational and self-management skills. These differences highlight how STEM programs tend to incorporate more reasoning-based and argumentation-based competencies, while BHASE programs prioritize organizational and interpersonal effectiveness.

Further analysis shows that STEM-related programs closely align with programs containing the keyword “computer”, with only minor variations in ranking after the top five skills. Similarly, BHASE-related programs align closely with agriculture-related programs, again with slight reordering in skill rankings (see Appendix, Figure 2). These patterns suggest that while foundational skills remain largely consistent across disciplines, subtle differences reflect domain-specific priorities.

Overall, both STEM and BHASE streams share a strong common core of foundational skills, particularly interpersonal communication, decision-making, active listening, and ethics. However, STEM programs place relatively greater emphasis on analytical and argumentation-based skills, whereas BHASE programs highlight organizational competencies alongside communication skills.

Appendix

Count	Key Observation	Runing Time
1	Generated BHASE, STEM, and All Program histogram plots; top skills (with similarity scores greater than 0.5) include Interpersonal Communication, Decision Making, Creativity, and Active Listening. Output saved successfully.	1 hour 10 min
2	Results reproducible; minor ranking changes but same dominant soft skills	1 hours 35 min
3	Soft skills dominate similarity scores (~0.60–0.70) across all programs; Similar top skills but in different rankings	1 hour 30 min
4	STEM adds strategic thinking, analytical, problem solving; overall similar to BASE	1 hour 30 min
5	“All programs” shows only minor ranking differences	1 hour 25 min
6	Top skills include leadership + core soft skills; outputs saved successfully	1 hour 20 min
7	Results remain consistent across runs, with the hybrid method producing similar rankings that highlight foundational skills, particularly interpersonal communication and active listening	1 hour 30 min

Figure 1: Summary of results across multiple runs of Hybrid NLP Analysis

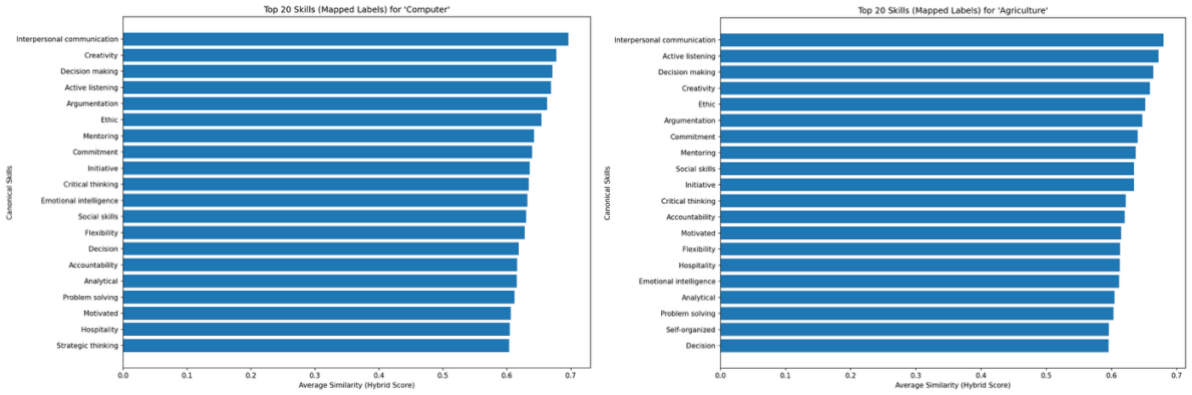


Figure 2: Top 20 skills for the program contains “Computer” versus programs contains “Agriculture” (case-insensitive)

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